

QIU-HAN GU

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Education

University of Illinois Urbana-Champaign

Ph.D. in Computer Science

Advisor: Prof. Gagandeep Singh

Aug.2026 – Present

Champaign, United States

University of Illinois Urbana-Champaign

Master of Science in Computer Science

Advisor: Prof. Gagandeep Singh

Aug.2024 – May.2026

Champaign, United States

Nanjing University

Bachelor of Science in Computer Science and Technology

Advisor: Prof. Linzhang Wang and Prof. Yu Wang

Sep.2020 – Jun.2024

Nanjing, China

Research Interests

I am broadly interested in the intersection of formal methods and machine learning. My current research focuses on probabilistic abstract interpretation, with applications to the verification and analysis of probabilistic programs and machine learning models. I am also interested in model synthesis for formal reasoning, exploring the complementary direction of using machine learning to assist formal methods.

Publications & Preprints

Qiuhan Gu, Gagandeep Singh, “*A Nonlinear Abstract Domain for Probabilistic Programs*” under submission.

Qiuhan Gu, Avaljot Singh, Gagandeep Singh, “*SAIL: Sound Abstract Interpreters with LLMs*” PLDI, 2026. [\[paper\]](#) [\[code\]](#)

Qiuhan Gu, “*LLM-Based Code Generation Method for Golang Compiler Testing*,” SRC@FSE (1st place), 2023. [\[paper\]](#) [\[code\]](#)

Honors & Awards

- **ESEC/FSE’23 Student Research Competition, 1st Prize (Undergraduate Division).** San Francisco, United States. Dec.2023.
- **Huawei Scholarship.** Nanjing University, China. Oct.2023.
- **National College Student Entrepreneurship Competition, 1st Prize.** China. Mar.2023.
- **National Scholarship of Undergraduate.** China. Dec.2021.

Research Experience

Probabilistic Abstract Interpretation

Research Assistant. Advisor: Prof. Gagandeep Singh

Jan.2026 – Now

FOCAL Lab, University of Illinois

- Developed Expohedron, a nonlinear abstract domain that represents probability distributions using interval-partitioned exponential-affine bounds, enabling sound and precise reasoning over continuous, unbounded, and heavy-tailed distributions.
- Derived sound abstract transformers for core, nonlinear probabilistic operations and extended the framework to verification of Bayesian neural networks.
- Led to a first-author submission to PLDI 2027.

Cost-Driven Synthesis of Sound Abstract Interpreters

Research Assistant. Advisor: Prof. Gagandeep Singh

Jun.2025 – Nov.2025

FOCAL Lab, University of Illinois

- Developed SAIL, the first unified framework to automatically synthesize globally sound abstract interpreters across multiple abstract domains.
- Formalized the synthesis process as a constrained optimization problem over an infinite search space of abstract transformers and introduced a novel mathematically grounded cost function that quantitatively measures unsoundness deviation to guide refinement.
- Empirically demonstrated the framework’s capability to match handcrafted transformer quality and synthesize globally sound transformers for complex nonlinear operators absent from the literature.
- Led to a first-author paper accepted at PLDI 2026.

Enhancing DL System Fuzzing via LLMs

Sep.2024 – Dec.2024

Intern. Advisor: Prof. Lingming Zhang and Prof. Darko Marinov

University of Illinois

- Developed an automated pipeline to generate and validate deep learning operator constraints using large language models, improving operator validity from 46.3% to 57.6%.
- Designed a domain-specific language to represent operator constraints and incorporated symbolic verification techniques to ensure correctness.
- Integrated LLM-generated operators into NNSmith and NeuRI fuzzing frameworks, achieving higher coverage and discovering multiple compiler bugs.

LLM-Based Code Generation Method for Golang Compiler Testing

Sept.2023 – Dec.2023

Undergraduate Researcher. Advisor: Prof. Yu Wang

SEG, Nanjing University

- Implemented a novel LLM-based high-quality code generation method for the Golang compiler, generating testcases with 3.38% average coverage and only 2.79% of them had syntax errors.
- Won 1st prize in the undergraduate division of the ESEC/FSE’23 Student Research Competition for a first-authored paper.

LLM-Based Code Classification Method (Bachelor Thesis)

Feb.2024 – Jun.2024

Undergraduate Researcher. Advisor: Prof. Yu Wang

SEG, Nanjing University

- Implemented an LLM-based high-quality code classification method for the Go language, achieving an accuracy of 96.4%.
- Combined large language models and traditional methods to efficiently model code in terms of lexical, syntactic, and semantic aspects.
- Defined custom classification categories, providing fine-grained datasets for downstream tasks.

Automatic Detection of Intracranial Aneurysms Based on Deep Learning

May.2022 – Apr.2023

Undergraduate Researcher.

Nanjing University

- Developed a clinically deployable deep learning system for automated intracranial aneurysm detection from CTA images and built a unified web platform for medical institutions to access and run the diagnostic system.
- Created a comprehensive implementation plan covering system design, deployment pipeline, and business integration.
- Won the first prize in National College Student Entrepreneurship Competition in 2023 (1/154).

Teaching Experience

- University of Illinois Urbana-Champaign: CS427 Software Engineering, 2025 Fall (*Teaching Assistant*)
- University of Illinois Urbana-Champaign: CS101 Introduction to Computer Science, 2025 Spring (*Teaching Assistant*)
- University of Illinois Urbana-Champaign: CS427 Software Engineering, 2024 Fall (*Teaching Assistant*)
- Nanjing University: Advanced Python Programming, 2024 Spring (*Teaching Assistant*)
- Nanjing University: Advanced Python Programming, 2023 Fall (*Teaching Assistant*)

Academic Service

Artifact Evaluation Committee: PLDI’26, POPL’27